

Avaliação 2 Cálculo Diferencial e Integral

1) $L(x) = -4x + 70$

$L(230) = -4 \cdot 230 + 70 = -930 \text{ reais}$

2) a) $\int \cos^2(3x) \sin(3x) dx$

$= \int u^2 \left(\frac{du}{-3} \right) = -\frac{1}{3} \int u^2 du$

$= -\frac{1}{3} \frac{u^3}{3} + C = \boxed{-\frac{1}{9} \cos^3(3x) + C}$

$u = \cos(3x)$

$du = -3 \sin(3x) dx$

$\frac{du}{-3} = \sin(3x) dx$

2) b) $\int (x^4 + x - 5) dx = \int x^4 dx + \int x dx - 5 \int dx$

$= \frac{x^5}{5} + \frac{x^2}{2} - 5x + C$

2) c) $\int \frac{1 + 2x^3 - x^2}{x^3} dx = \int \frac{dx}{x^3} + 2 \int \frac{x^3}{x^3} dx - \int \frac{x^2}{x^3} dx$

$= \int x^{-3} dx + 2 \int dx - \int \frac{dx}{x}$

$= \frac{x^{-2}}{-2} + 2x - \ln|x| + C$

$= \boxed{-\frac{1}{2x^2} + 2x - \ln|x| + C}$

$= -\frac{1}{2x^2} + 2x - \ln|x| + C$

$$3. a) \int 9x^2 \sin(x^3) dx, u = x^3$$

$$du = 3x^2 dx$$

$$\frac{du}{3} = x^2 dx$$

$$= 9 \int x^2 \sin(x^3) dx$$

$$= 9 \int \sin u \frac{du}{3} = \frac{9}{3} \int \sin u du = 3 \int \sin u du$$

$$= -3 \cos u + C = \boxed{-3 \cos(x^3) + C}$$

e
e
e
e

$$3. b) \int x \sqrt{x^2 + 6} dx \quad u = x^2 + 6$$

$$du = 2x dx$$

$$\frac{du}{2} = x dx$$

$$= \int u^{1/2} \frac{du}{2} = \frac{1}{2} \int u^{1/2} du$$

$$= \frac{1}{2} \cdot \frac{2}{3/2} u^{3/2} + C = \frac{1}{2} \cdot \frac{2}{3} u^{3/2} + C$$

$$= \boxed{\frac{1}{3} \sqrt{(x^2 + 6)^3} + C}$$

$$\frac{1}{2} + 1 = \frac{1+2}{2} = \frac{3}{2}$$

e
e
e
e

$$4. a) \int (4x-2)^6 dx \quad u = 4x-2$$

$$du = 4 dx$$

$$\frac{du}{4} = dx$$

$$= \int u^6 \frac{du}{4} = \frac{1}{4} \int u^6 du$$

$$= \frac{1}{4} \frac{u^7}{7} + C = \frac{1}{28} u^7 + C = \frac{1}{28} (4x-2)^7 + C = \boxed{\frac{(4x-2)^7}{28} + C}$$

e
e
e

$$4. b) \int t^2 e^{-t^3} dt$$

$$u = -t^3 \Rightarrow du = -3t^2 dt$$

$$\frac{du}{-3} = t^2 dt$$

$$= \int e^u \frac{du}{-3} = -\frac{1}{3} \int e^u du$$

$$= -\frac{1}{3} e^u + C = \boxed{-\frac{1}{3} e^{-t^3} + C}$$

e
e
e

$$4.c) \int \frac{\sec^2(\sqrt{x})}{\sqrt{x}} dx$$

$$= \int \sec^2 u \cdot 2 du$$

$$= 2 \int \sec^2 u du$$

$$= 2 \operatorname{tg} u + c$$

$$= 2 \operatorname{tg} \sqrt{x} + c$$

$$u = \sqrt{x} \quad \therefore u = x^{1/2} \quad du = \frac{1}{2} x^{-1/2} dx$$

$$du = \frac{1}{2\sqrt{x}} dx \quad \frac{1}{2} = \frac{1}{2} \cdot \frac{1-2}{2} = \frac{1}{2}$$

$$2 du = \frac{dx}{\sqrt{x}}$$

$$\begin{matrix} e & e \\ e & e \end{matrix}$$

$$5. a) \int (3x^2 + 2x) e^{3x} dx$$

$$u = 3x^2 + 2x$$

$$du = (6x + 2) dx$$

$$dv = e^{3x} dx$$

$$v = \int e^{3x} dx$$

$$= (3x^2 + 2x) \left(\frac{1}{3} e^{3x} \right) - \int \frac{1}{3} e^{3x} (6x + 2) dx$$

$$= \frac{1}{3} (3x^2 + 2x) e^{3x} - \frac{1}{3} \int (6x + 2) e^{3x} dx$$

$$\begin{matrix} u = 6x + 2 & dv = e^{3x} dx \\ du = 6 dx & v = \frac{1}{3} e^{3x} \end{matrix}$$

$$\begin{matrix} u = 3x \\ du = 3 dx \\ \frac{du}{3} = dx \end{matrix}$$

$$v = \int e^u \frac{du}{3}$$

$$v = \frac{1}{3} e^{3x} + c$$

$$= \frac{1}{3} (3x^2 + 2x) e^{3x} - \frac{1}{3} \left[(6x + 2) \cdot \frac{1}{3} e^{3x} - \int \frac{1}{3} e^{3x} 6 dx \right]$$

$$= \frac{1}{3} (3x^2 + 2x) e^{3x} - \frac{1}{9} (6x + 2) e^{3x} + \frac{6}{9} \int e^{3x} dx$$

$$= \frac{1}{3} (3x^2 + 2x) e^{3x} - \frac{1}{9} (6x + 2) e^{3x} + \frac{6}{9} \left(\frac{1}{3} e^{3x} \right) + c$$

$$= \frac{1}{3} (3x^2 + 2x) e^{3x} - \frac{1}{9} (6x + 2) e^{3x} + \frac{2}{3} e^{3x} + c$$

$$\begin{matrix} c \\ c \\ c \\ c \\ c \\ c \\ c \end{matrix}$$

$$5.b) \int x^4 \ln x^3 dx$$

$$u = \ln x^3$$

$$dv = x^4 dx$$

$$du = \frac{3x^2}{x^3} dx$$

$$v = \int x^4 dx$$

$$v = \frac{x^5}{5} + C$$

$$= \ln x^3 \cdot \frac{x^5}{5} - \int \frac{x^5}{5} \cdot \frac{3x^2}{x^3} dx$$

$$= \frac{x^5}{5} \ln x^3 - \frac{3}{5} \int x^4 dx = \frac{x^5}{5} \ln x^3 - \frac{3}{5} \frac{x^5}{5} + C$$

$$= \frac{x^5}{5} \ln x^3 - \frac{3}{25} x^5 + C$$

e
e
e
e

$$6. \int (2x+1) \sin x dx$$

$$u = 2x+1$$

$$dv = \sin x dx$$

$$du = 2 dx$$

$$v = \int \sin x dx$$

$$v = -\cos x + C$$

$$= (2x+1)(-\cos x) - \int (-\cos x) 2 dx$$

$$= -(2x+1) \cos x + 2 \int \cos x dx$$

$$= -(2x+1) \cos x + 2 \sin x + C$$

$$= -2x \cos x - \cos x + 2 \sin x + C$$

$$= 2(\sin x - x \cos x) - \cos x + C$$

$$7. f(x) = -x^3 + 8x$$

$$-2 \leq x \leq 4$$

$$f'(x) = 0$$

$$-3x^2 + 8 = 0$$

$$-3x^2 = -8$$

$$3x^2 = 8$$

$$x^2 = \frac{8}{3}$$

$$x = \pm \sqrt{\frac{8}{3}} = \pm 1,63$$

$$f(-2) = -(-2)^3 + 8(-2)$$

$$f(-2) = +8 - 16 = -8 = \text{max}$$

$$f(4) = -4^3 + 8 \cdot 4$$

$$f(4) = -64 + 32$$

$$f(4) = -32 = \text{min}$$